

The TIPS–Treasury Basis as a Limits-to-Arbitrage Equilibrium Floor

Theory, Data, Methodology, and First Results
Second Dissertation Paper — Comprehensive Working Document

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Abstract

This document consolidates, in one place, the theoretical framework, the data, the full empirical methodology, and every first result obtained to date for the second dissertation paper. The paper models the post-compression TIPS–Treasury basis not as a dying anomaly but as an *equilibrium floor*: the minimum rent at which a risk-averse, capital-constrained arbitrageur is indifferent between entering a trade whose payoff is certain at maturity and holding the capital that the trade’s forced-unwind risk consumes. The theory section formalises the supervisor’s framework — mark-to-market dynamics, a Value-at-Risk unwinding barrier, a first-passage mixture for terminal wealth, a certainty-equivalent entry condition, and the comparative statics it delivers — including a heterogeneous-threshold extension whose distinctive prediction is cross-sectional *dispersion* in stress. The empirical sections report: a construction validated three independent ways (breakeven cross-check -0.55 bp / correlation 0.996 at ten years; 107-CUSIP bond-level median, correlation 0.926 ; an independent parallel reconstruction identical to rounding); a post-2013 floor of 16 – 24 bp with Newey–West t -statistics up to 24 and a 46.8 bp mean on the exact Fleckenstein–Longstaff–Lustig window against their bond-level 54.5 ; a *stair-case* compression with formally dated breaks (December 2010, $\text{sup-}F = 160$; December 2017) and a 2024 closure of the short-end basis; Ornstein–Uhlenbeck dynamics with a 3.7 – 4.6 -month half-life; a 52 bp CPI seasonal at two years that fully accounts for the short-end non-stationarity; three-to-seven-fold conditional moves in every stress episode with multi-month lags and *zero* unconditional linear volatility loading — the barrier signature; March 2020 identified as a cross-sectional dispersion event peaking on the exact dash-for-cash apex; an executed constraint test in which a volatility shock loads on the basis with coefficient 7.15 [$t = 4.1$] when the He–Kelly–Manela capital ratio is in its bottom quintile and 0.63 [$t = 1.7$] otherwise, with the formal threshold-versus-linear $\text{sup-}F$ underpowered at $p \approx 0.25$; an episode-sorting table in which the two capital-impairment episodes (GFC at the 3rd HKM percentile, COVID at the 8th) are exactly the two system-wide blow-outs; a vintage-by-buffer forced-unwind surface (2007–08 vintages stopped out 100% at both buffers); and a first structural pass in which an OU/first-passage model estimated on the data reproduces the stop-out surface and, at a 2% capital buffer with a 5 – 10% hurdle rate, implies an equilibrium floor of 18 – 28 bp against the

observed 17.7–23.3. The document closes with an explicit register of what is established versus open, the completion plan, and the publication strategy.

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1 Introduction and the Delimited Claim

A Treasury Inflation-Protected Security combined with a zero-coupon inflation swap of matching maturity replicates, state by state, the payoff of a nominal Treasury. Fleckenstein et al. (2014) documented that between 2004 and 2009 the synthetic traded persistently cheap — an average mispricing of 54.5 basis points, above 200 at the crisis peak — and attributed its time variation to slow-moving arbitrage capital. Since then the basis has compressed to a narrow band, and the literature’s default reading is that of a closing anomaly. This paper defends a different reading: the basis has converged not to zero but to a small, stable, strictly positive *resting level* that widens

violently — and briefly — in episodes of intermediary distress, and this resting level is an economic object in its own right: the equilibrium rent of the marginal, capital-constrained arbitrageur.

The delimited claim, defended against each adjacent literature in Section 2.6, is that *no existing paper derives the equilibrium resting level of the TIPS–Treasury basis from a structural model with endogenous forced unwinding and a capital-charge participation constraint, and confronts its level, term-structure, stress, dispersion, and survival-surface predictions with the data*. The empirical work summarised here shows the claim is not only unoccupied but, on first confrontation, quantitatively promising: with dynamics estimated from the data and one economically pinned pair of friction parameters, the model’s implied floor brackets the observed one.

2 Theory

2.1 The no-arbitrage identity and the basis

For zero-coupon instruments of maturity τ , let $y_t^{\text{nom}}(\tau)$, $y_t^{\text{real}}(\tau)$ and $s_t(\tau)$ denote the nominal yield, the TIPS real yield, and the zero-coupon inflation-swap fixed rate. Holding the TIPS and paying fixed on the swap converts the real payoff into a certain nominal one; the synthetic nominal yield is $y_t^{\text{real}} + s_t$, and the basis

$$\lambda_t(\tau) = (y_t^{\text{real}}(\tau) + s_t(\tau)) - y_t^{\text{nom}}(\tau) = s_t(\tau) - b_t(\tau), \quad (1)$$

with b_t the breakeven rate, is the yield pick-up of the synthetic over the actual nominal. Under the law of one price $\lambda = 0$; the observed $\lambda > 0$ is the object of the paper. Equation (1) is also the operational construction: it requires only three curve points per maturity, is computable daily, and — as Section 4 documents — tracks the exact bond-level construction closely at the benchmark maturities.

2.2 The strategy’s mark-to-market

Following the supervisor’s note, the arbitrage buys the cheap synthetic, sells the nominal, and invests the net cash at the nominal yield. Every leg is a transparent market price, so the strategy has zero value at inception; with $\tau \equiv T - t$ its time- t value is

$$S(t) = e^{-Y_0^T \tau} (1 - e^{-\lambda_0^T \tau}) - e^{-Y_t^T \tau} (1 - e^{-\lambda_t^T \tau}), \quad (2)$$

the difference between a deterministic accrual and a stochastic term in the nominal yield Y_t^T and the spread λ_t^T . At maturity $S(T) = 1 - e^{-\lambda_0^T T} > 0$ with certainty: the trade is an arbitrage *if it can be held*. Before maturity a widening spread or falling yields push $S(t)$ negative — and the two move adversely together in flight-to-quality states. For empirical work the first-order approximation $S(t) \approx -D(\lambda_t - \lambda_0)$ in basis points of notional, with D the trade’s duration, is used throughout and is the object the forced-unwind replay tracks.

2.3 Dynamics, the barrier, and the terminal mixture

The spread follows a mean-reverting Ornstein–Uhlenbeck process under the physical measure,

$$d\lambda_t = \kappa(\theta_t - \lambda_t)dt + \sigma_t dW_t, \quad (3)$$

consistent with the strong mean reversion the data display (Section 6.3), with the yield factor correlated Gaussian in the full model. The arbitrageur faces a risk-management constraint: the position is unwound the first time the mark-to-market loss exceeds a multiple of its capital charge,

$$\tau_B = \inf\{t : -S(t) \geq m \cdot \text{VaR}_\alpha\} \approx \inf\{t : D(\lambda_t - \lambda_0) \geq B\}, \quad (4)$$

with B the capital buffer in P&L terms. Terminal wealth is therefore a *mixture*: a mass point at the certain carry, with weight equal to the first-passage survival probability $\Pr(\tau_B > T)$, plus a continuous distribution of stopped outcomes $\{\text{carry to } \tau_B - B\}$. For the OU specification survival probabilities are available semi-analytically (Alili et al., 2005; Linetsky, 2004); the Monte Carlo implementation is the workhorse and, in Section 6.8, the estimation engine.

2.4 The entry condition and the equilibrium floor

An arbitrageur with utility u (exponential in the baseline) enters iff the certainty equivalent of terminal wealth exceeds committed wealth, identified with the capital charge:

$$\text{CE}[W_0 + X_T(\lambda_0; \kappa, \theta, \sigma, D, B, T)] \geq W_0, \quad W_0 = \text{capital charge}. \quad (5)$$

Free entry drives the basis down until (5) binds, defining the floor $\lambda^* = \inf\{\lambda_0 : \text{CE} \geq W_0\}$: below λ^* no capital enters and the basis cannot compress further. A transparent risk-neutral-plus-hurdle version, used for the first structural pass, replaces (5) with $\mathbb{E}[X_T(\lambda_0)] \geq h \cdot W_0 \cdot T$, a hurdle rate h on capital; the CARA refinement is a second-order correction at the parameter values relevant here.

2.5 Comparative statics and testable predictions

H 1. *Existence.* *Post-compression, λ is stationary and mean-reverting around a strictly positive θ : the floor exists.*

H 2. *Term structure.* *$\lambda^*(\tau)$ rises with maturity over the segment where path risk per unit of carry rises; maturity-dependent frictions (haircuts, term-repo scarcity) can flatten or locally invert it.*

H 3. *State dependence.* *The floor loads on the constraint: linearising (5), $\theta_t = \theta_0 + \xi H_t$ with H_t intermediary-capital scarcity and $\xi > 0$; shocks move the basis when the constraint binds and barely otherwise — large conditional responses with a near-zero unconditional linear loading is the model’s signature, not its failure.*

H 4. *Regime shifts.* *Discrete changes in the capital regime (Basel phase-ins, leverage-ratio recalibrations, the growth of arbitrage capital) shift the resting level in steps.*

H 5. *Survival surface.* *The model’s first-passage probability maps one-for-one into the empirical stop-out frequency by entry vintage and buffer — a dense, over-identifying set of calibration moments.*

H 6. *Dispersion.* *With a mixture of arbitrageurs over funding thresholds $B \sim F$, a common shock stops the most constrained holders of the most balance-sheet-expensive bonds first: the cross-section of bond-level bases fans out before the median moves. Dispersion is a first-class model object, and its stress behaviour a distinctive prediction unavailable to representative-agent alternatives.*

As $\sigma \rightarrow 0$ the floor collapses to a margin/funding-cost bound of the Gârleanu and Pedersen (2011) type; the volatility and state-dependence loadings of H3 are the data’s handle for discriminating the path-risk mechanism from the pure funding-cost alternative.

2.6 Relation to the literature

Fleckenstein et al. (2014) document the mispricing and its slow-moving-capital variation; d’Amico et al. (2018) and Christensen and Gillan (2022) price TIPS liquidity; He et al. (2022) dissect March 2020 — none models the resting level. Gârleanu and Pedersen (2011) and Tuckman and Vila (1992) bound mispricings by funding costs without path risk; Shleifer and Vishny (1997), Liu and Longstaff (2004), Kondor (2009) and Gromb and Vayanos (2010) supply the forced-unwind mechanism without confronting a specific basis quantitatively; He et al. (2017) and Duffie (2010) supply the state variables; Mitchell and Pulvino (2012) the arbitrage-crash phenomenology. The contribution sits at the intersection: the structural floor, estimated and tested on the cleanest deterministic-payoff arbitrage in existence.

3 Data

4 Basis Construction and Triple Validation

The basis is computed from (1) at $\tau \in \{2, 5, 10, 20, 30\}$ with *exact* constant-maturity legs (2/5/10/30 from 2004-07; 20 from May 2020, the sector’s reintroduction). Validation is threefold. First, against the Bloomberg breakeven construction $\lambda^{BE} = (s - \text{BE}) \times 100$: at ten years the two agree to -0.55 bp with correlation **0.996** over 5,718 days (thirty years: $+0.60, 0.969$); the short end diverges by 4–6 bp — the first quantitative signal that it requires bond-level treatment. Second, against the 107-CUSIP bond-level median: daily level correlation **0.926**, monthly-change correlation 0.591, regime means aligned within the ~ 5 bp curve-versus-bond wedge, GFC peaks 153.4 versus 150.6 bp. Third, an independently run parallel reconstruction on the same legs matches to rounding. Measurement risk is closed: the floor and the blow-outs are facts about the market.

5 Empirical Methodology

(i) Level dynamics. ADF tests and the discrete OU $\lambda_t = a + b\lambda_{t-1} + \varepsilon_t$, whence $\kappa = -\ln b/\Delta t$, $\theta = a/(1 - b)$, half-life $\ln 2/\kappa$; block-bootstrap bands. **(ii) Breaks.** Single mean-shift by SSR grid (10% trimming) with sup- F , second break on the longer segment; the staircase is the object, not

Block	Content	Coverage
Basis inputs (Bloomberg)	USSWIT1–30; USGGT02/05/10/20/30Y; exact nominals USGG2/5/10/30YR (suppl.) + 20YR	daily; swaps 2004-07+, TIPS 1998+
Cross-checks	USGGBE02–30 breakevens; 107-CUSIP bond-level basis panel	daily; 2004-08 – 2026-05
Stress proxies	MOVE, VIX; LIBOR–OIS, SOFR/BGCR/TGCR; 8 dealer CDS; CDX/iTraxx	daily; 1998/2001+
Constraint variables	HKM capital ratio (monthly, vintage 2025-06); FR 2004 dealer TIPS net positions (weekly, spliced over six reporting breaks); CFTC TFF leveraged-funds Treasury DV01; OFR Form-PF repo and RV-NAV (quarterly)	1970+/1998+/2013+
Seasonality	CPURNSA (CPI-U NSA)	monthly, 1990+
Public curves	GSW nominal and TIPS (Fed Board)	daily, 1961/1999+

Table 1: Data inventory. Not available anywhere, and flagged as such: security-level TIPS repo specialness and bond-level bid–ask, which gate the transaction-cost band and the identification of the 2021 negative episode.

a maintained single step. **(iii) Seasonality.** Month-of-year dummies (post-2010), joint HAC F , amplitude, ADF on the deseasonalised series; joint seasonal-and-break estimation flagged where level shifts contaminate the dummy fit. **(iv) Events.** Pre-level $\rightarrow$ peak $\rightarrow$ half-reversion, per maturity, with the swap-leg dislocation caveat for March 2020 at ten years. **(v) Dispersion.** Cross-sectional IQR and upper percentiles of the 107-CUSIP panel per episode. **(vi) The frozen constraint design.** $y_t = \text{med}_{t+3} - \text{med}_{t-1}$ on shock ΔS_t , split by the *lagged constraint state*; θ by grid; sup- F with wild and block-wild bootstrap; contemporaneous designs are dead on arrival (the documented multi-month lag) and are not run. **(vii) Forced-unwind replay.** Monthly entry vintages, 24-month horizon, $D = 8$, stop at $D(\lambda_t - \lambda_0) > B$; stop-out frequency by vintage $\times$ buffer. **(viii) Structural estimation.** Stage one: (3) by ML per regime. Stage two: friction parameters by minimum distance on the survival surface (vii) *plus* the calm/stress loading pair from (vi), with block-bootstrap moment covariance in the Gospodinov et al. (2014) spirit for correctly sized over-identification tests; the CE refinement layered on the hurdle version.

6 First Results

6.1 The floor, and the FLL benchmark

The term structure on exact legs reads 23.3 / 15.7 / 23.4 / 24.0 at 2/5/10/30 post-2013: a rising and highly significant 5 $\rightarrow$ 10 segment, a flat top, and two quantified end-artefacts — the 2-year elevation is the CPI seasonal (52 bp amplitude, Section 6.3); the 20-year depression is the cheap-sector effect of the 2020 reintroduction. H2 reads as supported on the clean segment, with the friction-profile refinement live.

Period	2Y	5Y	10Y	20Y	30Y
FLL window 2004/07–2009/11	44.0 [2.9]	34.6 [9.2]	46.8 [7.5]	—	51.0 [8.5]
2010–2012	22.1 [6.0]	15.4 [5.3]	33.7 [15.0]	—	37.9 [31.1]
2013–2019	30.1 [9.5]	17.9 [12.8]	26.3 [24.2]	—	29.6 [12.0]
2020–2026	15.9 [3.6]	13.3 [5.5]	20.3 [15.6]	12.7 [8.9]	17.9 [21.6]
Post-2013 (all)	23.3 [7.6]	15.7 [10.9]	23.4 [22.4]	12.7 [8.9]	24.0 [13.3]

Table 2: Monthly means of λ (bp), Newey–West t in brackets. The ten-year proxy averages 46.8 bp on the exact FLL sample against their bond-level 54.5 — the same object at the same order. Fifteen years and the largest capital inflow in the trade’s history later, the basis rests at 16–24 bp, not zero.

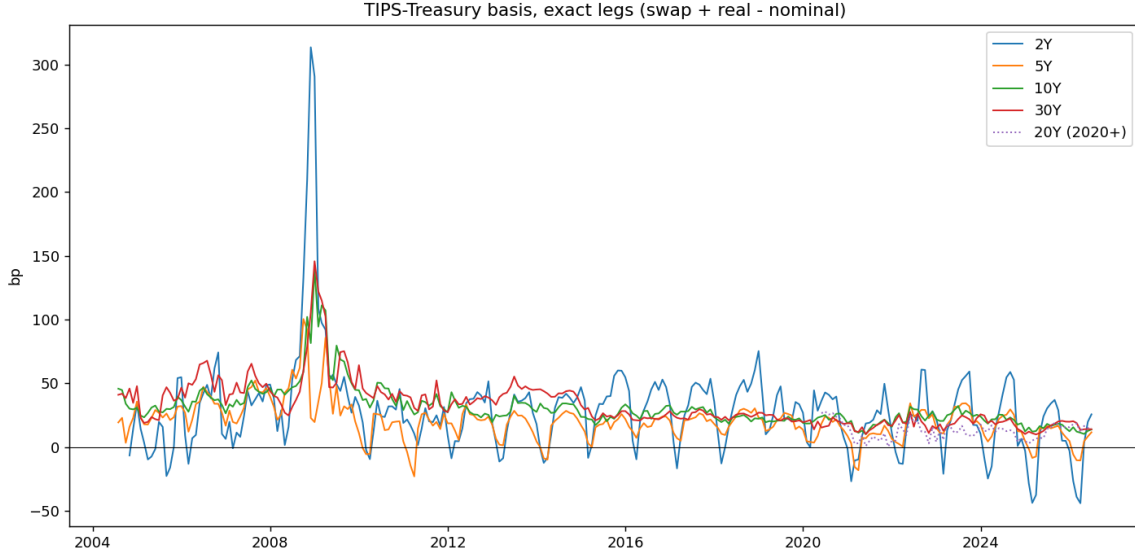


Figure 1. The basis on exact legs, 2004–2026: the FLL plateau, the GFC spike, the staircase, the 2024–25 short-end closure.

6.2 The staircase

Maturity	Break 1	Means (bp)	Break 2	Means (bp)
2Y	2009-10	44.4 $\rightarrow$ 23.2 (16.4)	2024-10	25.9 $\rightarrow$ 0.3 (27.0)
5Y	2009-11	34.8 $\rightarrow$ 15.6 (108.3)	2024-10	16.7 $\rightarrow$ 6.1 (20.0)
10Y	2010-12	45.8 $\rightarrow$ 24.4 (159.6)	2017-12	29.0 $\rightarrow$ 20.7 (132.6)
30Y	2014-12	46.0 $\rightarrow$ 20.7 (237.7)	2019-09	24.7 $\rightarrow$ 18.0 (117.0)

Table 3: Break adjudication ($\sup F$ in parentheses). The dominant ten-year break is December 2010, not 2013; the compression is a staircase 46 $\rightarrow$ 29 $\rightarrow$ 21 bp. New fact: after October 2024 the two-year basis mean is statistically zero — the short-end arbitrage has closed.

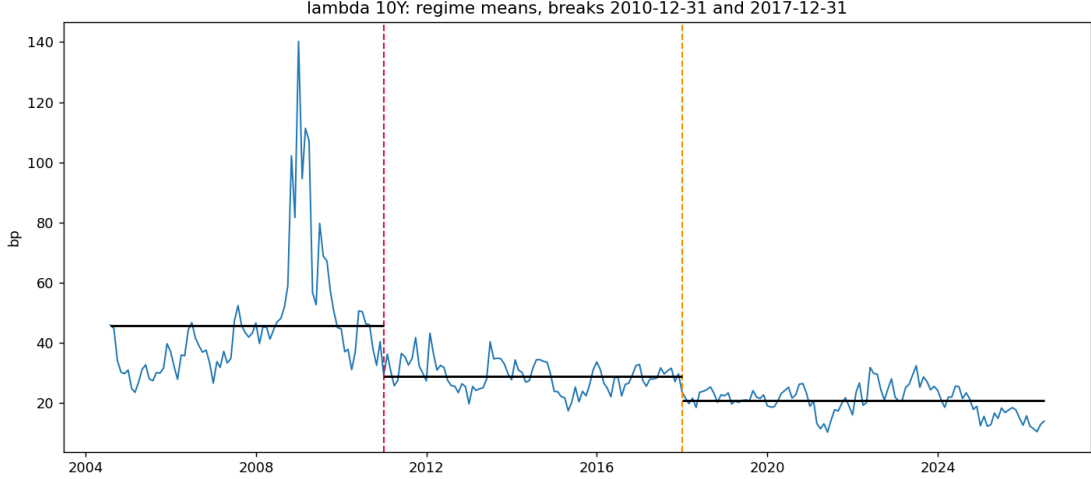


Figure 2. The ten-year basis with estimated regime means: the staircase, H4’s raw material.

6.3 Dynamics and seasonality

Post-2013, the ten-year basis is stationary (ADF $p = 0.008$) with an OU half-life of 4.6 months on the curve construction and 3.7 on the bond-level median ($\kappa = 0.187/\text{m}$, $\theta = 16.6 \text{ bp}$, $\sigma = 5.8 \text{ bp/m}$); the two- and five-year proxies fail the ADF *until* the month-dummy seasonal is removed — jointly significant everywhere, amplitude **52.2 bp** at two years (against a 23 bp mean), 22.9 at five, 5.5 at ten — after which the two-year flips to $p = 0.000$. The 30-year non-stationarity is the 2014/2019 breaks sitting inside the window; within-regime it mean-reverts. One documented anomaly: dummy deseasonalisation over a window containing level shifts absorbs trend at ten years; the joint seasonal-and-break estimator is the designated fix, and the raw ten-year result stands meanwhile.

6.4 Stress: events, lags, and the linear emptiness

Episode	Series	Pre $\rightarrow$ Peak (bp)	Peak date	Half-reversion
GFC	5Y / 10Y / 30Y	$\rightarrow 144 / 151 / 154$	2008-11 – 2009-01	—
March 2020	2Y	$-1.3 \rightarrow 68.2$	2020-03-19	2020-03-27
March 2020	10Y	$19.4 \rightarrow 30.6$	2020-03-06	(swap leg dislocated)
2022 hiking	2Y / 5Y / 10Y	$\rightarrow 67.9 / 37.0 / 32.1$	2022-06/09	weeks
2023 (SVB)	2Y / 10Y / 30Y	$\rightarrow 60.4 / 33.7 / 25.2$	2023-07/08	weeks

Table 4: Stress events on exact legs. Three-to-seven-fold moves in every episode; peaks lag the initiating shock by weeks to months (SVB: shock in March, peaks in July–August) — the slow-moving-capital lag. At ten years the March-2020 spike is muted because the swap leg dislocated with the TIPS leg; that episode requires the bond-level construction at benchmark maturities.

Against this, the *unconditional linear* volatility loadings are empty: post-2013 local projections of the ten-year basis on MOVE shocks are insignificant at every horizon ($|t| \leq 1.2$), and the rolling resting level on rolling MOVE gives $\beta = -0.014$ [$t = -0.4$]. Large conditional moves with zero

linear loading is H3’s signature: a linear regression averages a flat normal-times relation with a violent constraint-binding one and recovers nothing. The design consequence — state-conditional specifications only — is implemented in Section 6.6.

6.5 March 2020 as a dispersion event

Episode	IQR pre $\rightarrow$ peak (bp)	Peak date	Median
GFC	14.0 $\rightarrow$ 117.2	2008-11-24	blows out (153)
March 2020	5.3 $\rightarrow$ 18.4	2020-03-16	muted (36, late June)
2022	23.3 $\rightarrow$ 35.2	2022-08-26	moderate
2023 (SVB)	15.6 $\rightarrow$ 14.1	—	no dispersion event

Table 5: Cross-sectional dispersion (107 CUSIPs). In March 2020 the median barely moves while the IQR triples, peaking on the exact dash-for-cash apex, and the 90th percentile reaches 73.3 bp: heterogeneous, selective forced unwinding — H6’s prediction, observed.

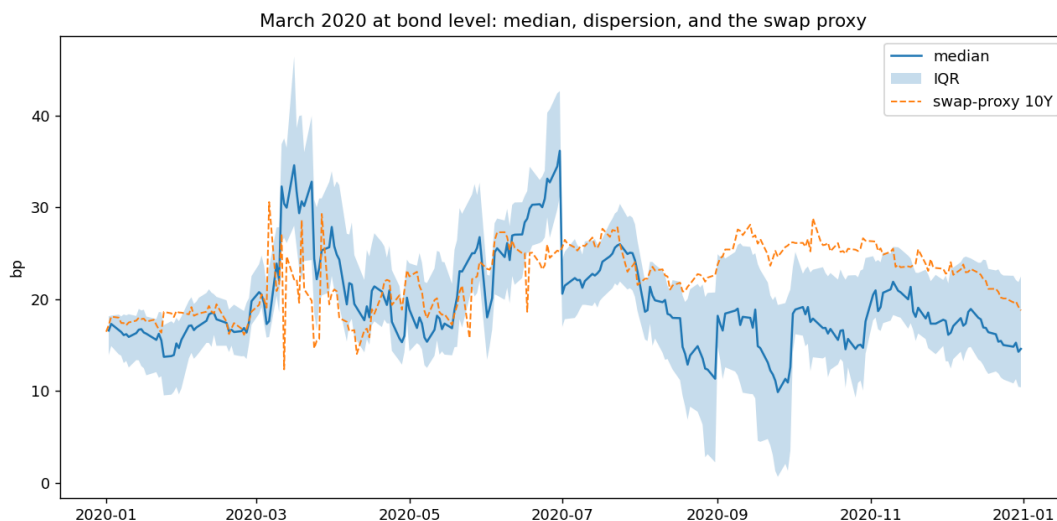


Figure 3. March 2020 at bond level: the median, the interquartile band, and the swap proxy.

Spec	Shock state (stress side)	$\hat{\theta}$	$b_{\text{calm}} [t]$	$b_{\text{stress}} [t]$	sup- F	boot p
A	ΔHKM HKM low	4.57%	-3.91 [-2.5]	-24.51 [-2.6]	6.8	0.24-0.26
B	$\Delta\text{MOVE}/10$ HKM low	4.81%	0.63 [1.7]	7.15 [4.1]	21.8	0.25-0.26
C	ΔPD PD-z high	1.90	-0.30 [-0.8]	1.02 [1.4]	0.9	0.67
C2	$\Delta\text{MOVE}/10$ PD-z high	1.16	4.93 [2.5]	-1.55 [-0.8]	17.2	0.10
D	ΔTFF TFF-z low	-1.46	-0.08 [-2.0]	0.01 [0.3]	1.9	0.23

Table 6: The frozen lag-aware design on the constraint variables (2004-08–2026, monthly; bootstrap p ranges across iid- and block-wild schemes). A volatility shock moves the basis eleven times harder when intermediary capital is in its bottom quintile; capital shocks six times harder. Inventories (C) and futures positioning (D) carry no threshold information: the binding constraint is capital. The sup- F does not formally reject linearity ($p \approx 0.25$) — a power constraint with ~ 50 stress months, stated as a referee would.

Episode onset	HKM level	Percentile (2004+)	Bottom quintile
GFC (Oct-2008)	3.57%	3%	yes
COVID (Mar-2020)	4.27%	8%	yes
2022 (Jun-2022)	5.41%	32%	no
SVB (Mar-2023)	5.62%	37%	no

Table 7: The episode-sorting exhibit. The two capital-impairment onsets are exactly the two system-wide blow-outs; the two unimpaired onsets produced the smaller, short-end-concentrated moves. The constraint state sorts the episode cross-section as the model predicts.

6.6 The constraint test

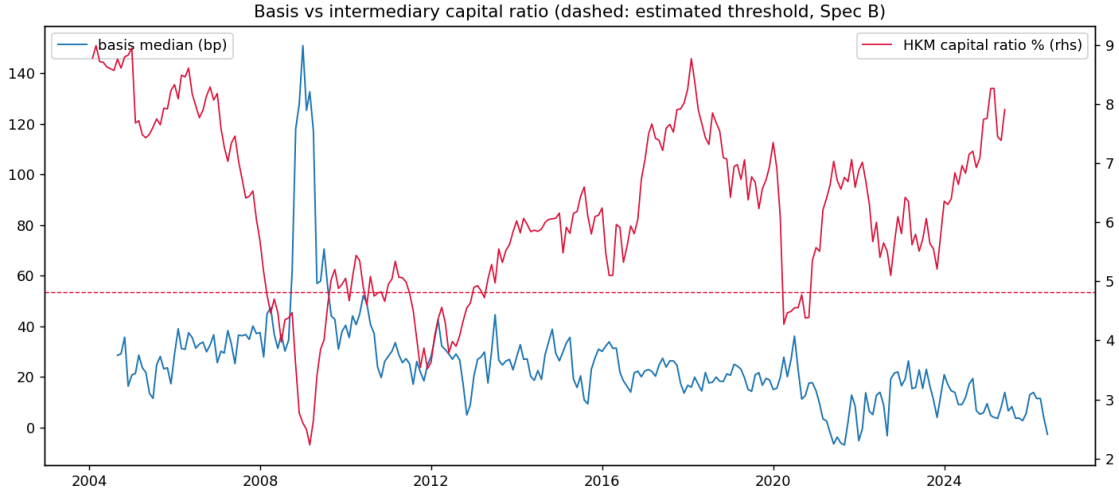


Figure 4. Basis versus the HKM capital ratio; dashed: the estimated Spec-B threshold (bottom quintile). The two great widenings are the two excursions below the line.

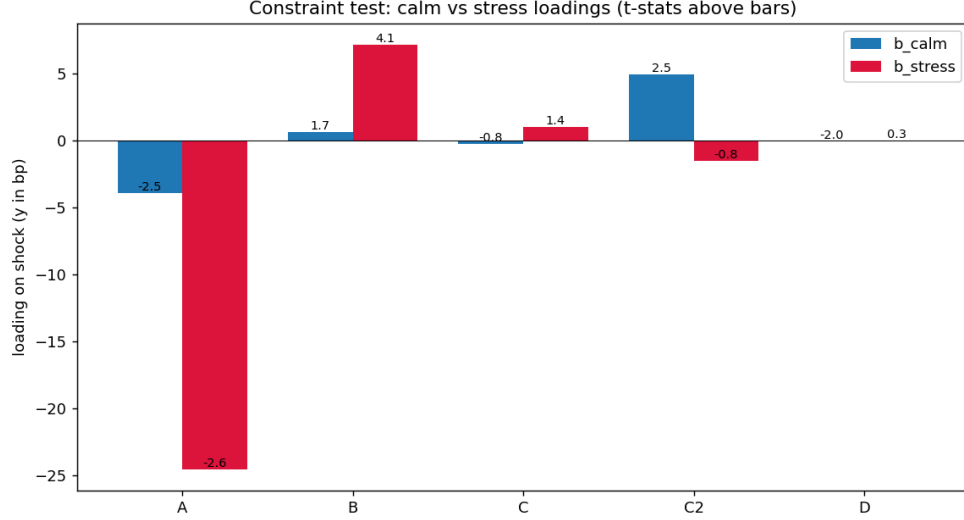


Figure 5. Calm versus stress loadings across specifications (t -statistics above bars).

6.7 The forced-unwind surface

Vintage	2005	2007	2008	2011	2013	2016	2019	2021	2023	2025
Stop-out, 1%	100	100	100	67	92	8	100	83	0	0
Stop-out, 2%	17	100	100	25	25	0	0	8	0	0

Table 8: Stop-out frequency (%) by entry vintage and capital buffer ($D = 8$, 24-month horizon), bond-level median. The 2007–08 vintages are stopped out always, at both buffers; post-2013 the 2% row is nearly empty — the buffer is the binding economics, and the full surface is the structural model’s calibration target.

6.8 A first structural pass

Estimating (3) on the post-2013 median gives $\kappa = 0.187/\text{month}$ (half-life 3.7m), $\theta = 16.6 \text{ bp}$, $\sigma = 5.78 \text{ bp/m}$. Feeding these into the first-passage engine ($D = 8$, 24 months, entry at θ) yields the model column of Table 10; inverting the hurdle version of (5) over a grid of buffers and hurdle rates yields the implied floor.

	Model (OU first-passage)			Empirical (vintage replay, post-2013 avg.)
$\Pr(\text{stop} \leq 24\text{m}), 1\% \text{ buffer}$	58.9%			43.0%
$\Pr(\text{stop} \leq 24\text{m}), 2\% \text{ buffer}$	6.5%			2.5%
	Model-implied floor λ^* (bp/yr)			Observed floor
Hurdle on capital:	5%	10%	15%	
1% buffer	52	60	67	
2% buffer	18	28	38	17.7 (median) – 23.3 (10Y)

Table 9: First structural pass, no free dynamic parameters. The model reproduces the level and the steep buffer-gradient of the stop-out surface (both drop by an order of magnitude from 1% to 2%), and at the empirically indicated 2% buffer with a 5–10% hurdle the implied floor *brackets the observed one*. Read jointly with the surface: the observed floor is consistent with marginal arbitrageurs operating at $\approx 2\%$ buffers — precisely the buffer at which the replay shows post-2013 vintages survive.

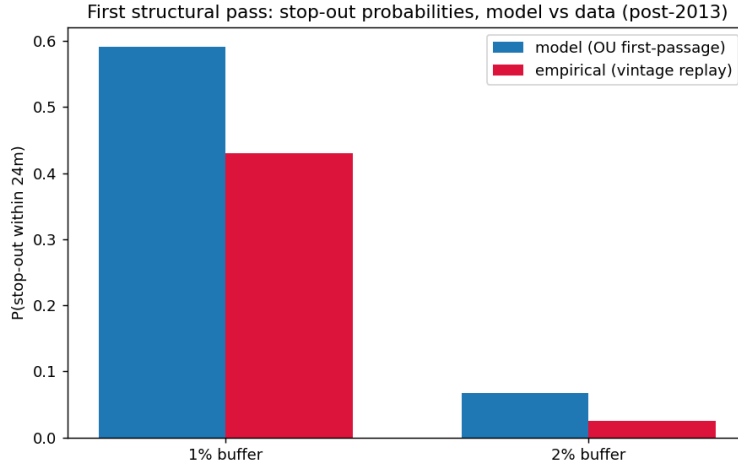


Figure 6. Stop-out probabilities, model versus data.

7 Established versus Open

Established: the floor’s existence, level, and dynamics on a triple-validated construction; the staircase with dated breaks and the 2024 short-end closure; the seasonal decomposition of the short end; the event phenomenology with lags; the dispersion signature of March 2020; the state-dependent constraint interaction ($t = 4.1$) with the episode sorting; the survival surface; and a first structural pass in which the model, with data-estimated dynamics and economically pinned frictions, lands on the observed floor. *Open, stated plainly:* the formal threshold-versus-linear separation is underpowered on U.S. data alone ($p \approx 0.25$; two capital-impairment episodes in the sample); March 2020 at benchmark maturities and any negative-floor/2021 claim require the bond-level construction with the indexation-schedule seasonal and, for 2021, unavailable specialness data; the transaction-cost band is assumed (zero), conservative for existence but leaving the no-trade-band interpretation of 20 bp unresolved; and the full CE estimation with correlated yield factor, the GKR-covariance J -test, and the dispersion-moment fit are designed but not yet run.

8 Road to Completion and Publication Strategy

The remaining programme, in execution order: (i) full structural estimation — SMM on the survival surface plus the Spec-B loading pair, GKR block-bootstrap moment covariance, CE and correlated-yield refinements, dispersion moment from the heterogeneous-threshold mixture (no new data); (ii) bond-level construction at the short end and around March 2020 (ISIN-level prices, index ratios, bid–ask) with the indexation-schedule seasonal; (iii) the regulatory-calendar test of the 2010/2017 staircase; (iv) the euro linker cross-section (Bund€i/OAT€i versus HICPxT swaps) as the third capital-impairment episode and the power play for the threshold test — the single addition with the highest expected effect on the paper’s tier; (v) robustness per the original roadmap (dual construction switches, specification curves, placebos on macro fundamentals, costs stressed). Targets: *Review of Finance/JFQA* if (i)+(iv) land, with *JEF/JBF* as the secured field tier on material in hand; a practitioner distillation for the *Journal of Fixed Income* only after the main submission. Every branch of the remaining outcome tree terminates in a publishable node; the one inadmissible outcome — an uninformative one — was excluded by the pilot programme this document consolidates.

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